

Basic Ciphers: A Beginner's Guide

1. ATBASH Cipher

Concept

The Atbash cipher is a substitution cipher where: - First letter becomes last letter (A Z) - Second letter becomes second-to-last letter (B Y) - And so on...

Conversion Table

Plain:	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher:	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

Example 1: Simple Word

Plaintext: HELLO

H → S (H is 8th from start, S is 8th from end)
E → V (E is 5th from start, V is 5th from end)
L → O (L is 12th from start, O is 12th from end)
L → O (L is 12th from start, O is 12th from end)
O → L (O is 15th from start, L is 15th from end)

Ciphertext: SVOOL

Example 2: Sentence

Plaintext: MEET AT NOON

M → N
E → V
E → V
T → G
A → Z
T → G
N → M
O → L
O → L
N → M

Ciphertext: NVVG ZG MLLM

Practice Problems - Atbash

1. Encrypt: CRYPTO
2. Decrypt: XRKSVI
3. Encrypt: KEEP IT SECRET

Solutions: 1. CRYPTO → XIBKGL 2. XRKSVI → CIPHER 3. KEEP IT SECRET → PVVK RG HVXIVG

2. ROT13 Cipher

Concept

- Rotate each letter 13 positions forward
- Because 13 is half of 26, applying ROT13 twice returns the original text

Quick Reference

Plain:	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
ROT13:	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M

Example 1: Simple Word

Plaintext: HELLO

H + 13 = U

E + 13 = R

L + 13 = Y

L + 13 = Y

O + 13 = B

Ciphertext: URYYYB

Example 2: With Wrap-around

Plaintext: PYTHON

P + 13 = C (P=16, 16+13=29, 29-26=3=C)

Y + 13 = L (Y=25, 25+13=38, 38-26=12=L)

T + 13 = G

H + 13 = U

O + 13 = B

N + 13 = A

Ciphertext: CLGUBA

Practice Problems - ROT13

1. Encrypt: CODE
2. Decrypt: PERNX
3. Encrypt: TOP SECRET

Solutions: 1. CODE → PBQR 2. PERNX → BREAK 3. TOP SECRET → GBC FRPERG

3. Morse Code

Concept

- Represents letters using dots (.) and dashes (-)
- Space between letters: 3 units
- Space between words: 7 units

Common Morse Code Table

A: .-	B: -...	C: -.-.	D: -..
E: .	F: ..-.	G: --.	H:
I: ..	J: .---	K: -. -	L: .-..
M: --	N: -.	O: ---	P: .--.
Q: --.-	R: .-.	S: ...	T: -
U: ..-	V: ...-	W: .--	X: -...-
Y: -.-	Z: --..		

Example 1: Simple Word

Plaintext: SOS

S: ...

O: ---

S: ...

Morse code: ... — ...

Example 2: Short Sentence

Plaintext: HELP ME

H:

E: .

L: . - . .

P: . - - .

[space]

M: - -

E: .

Morse code: -.-. -.-. - .

Practice Problems - Morse Code

1. Encode: HI
2. Decode: -.-. .
3. Encode: MAYDAY

Solutions: 1. HI $\rightarrow$ 2. -.-. . $\rightarrow$ HERE 3. MAYDAY $\rightarrow$ - .- -.- -.. .- -.-

General Tips for Teaching

1. Start with short words
2. Progress to longer words
3. Finally move to sentences
4. Always show the conversion process step by step
5. Encourage students to create their own examples
6. Use mnemonics where helpful

Common Student Mistakes to Watch For

1. Atbash: Forgetting to count positions correctly
2. ROT13: Incorrect handling of wrap-around
3. Morse: Confusion between similar patterns (e.g., E and I)

Practice Methods

1. Use flashcards for Morse code
2. Create lookup tables for Atbash
3. Mental math exercises for ROT13
4. Group exercises where students encrypt messages for each other
5. Validation by decrypting the encrypted message